# VES Institute of Technology, Mumbai
# Department of Computer Engineering

**Cryptocurrency Blockchain Development - Tutorial (2025-26) / Sem 6**

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1. A smart contract execution consumes 85,000 gas units. Gas price = 40 Gwei.
   If 1 ETH = $2500, find the transaction fee in ETH and USD.
   Hint : 1 gwei = $10^{-9}$ ETH

**Solution:** Total Gas Fee (ETH) = $85{,}000 \times 40 \times 10^{-9} = 3.4 \times 10^{-3}$ ETH
Fee in USD = $3.4 \times 10^{-3} \times 2500 = \$8.50$

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2. A blockchain network adjusts difficulty based on the time to mine 2016 blocks. Target time = 2 weeks = 14 days. If miners took 10 days to mine 2016 blocks, find the new difficulty. Given old difficulty = $1.0 \times 10^{12}$. Also, the formula to compute Difficulty is as follows:

$$\text{New Difficulty} = \text{Old Difficulty} \times \frac{\text{Actual Time}}{\text{Target Time}}$$

**Solution:**

$$\text{New Difficulty} = 1.0 \times 10^{12} \times \frac{10}{14} = 0.714 \times 10^{12}$$

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3. T1, T2, T3, T4 and T5 and the Transactions in the blockchain. Design the Merkle Tree. If transaction T3 is altered after the block is created, how would this affect the Merkle Root? How does this help maintain blockchain data integrity?

**Solution:**

Design a Merkle Tree for 5 transactions: T1, T2, T3, T4, and T5.

1. Hash each transaction: Calculate the hash of each individual transaction — H1 = hash(T1), H2 = hash(T2), H3 = hash(T3), H4 = hash(T4), and H5 = hash(T5).
2. Pair and hash leaf nodes: Since we have an odd number of transactions (5), duplicate the last hash H5 to make it even.
   Pairs to hash at first level:
   - Pair 1: H1 and H2 → H12 = hash(H1 + H2)
   - Pair 2: H3 and H4 → H34 = hash(H3 + H4)
   - Pair 3: H5 and H5 → H55 = hash(H5 + H5)
3. Next level: Now, hash pairs of these new hashes:
   - Pair A: H12 and H34 → H1234 = hash(H12 + H34)
   - Pair B: H55 duplicated again to make an even number → H5555 = hash(H55 + H55)
4. Root hash: Finally, hash H1234 and H5555 to get the Merkle Root:
   - Merkle Root = hash(H1234 + H5555)

**Analysis of Change:**

1. Change in T3's Hash: If transaction T3 changes, its hash $H_3 = hash(T3)$ changes to a new hash $H_3^{'}$.

2. Propagation of Changes Up the Tree: This altered hash changes the combined hash of its pair (here with T4's hash $H_4$) to $H'_{34} = hash(H'_3 + H_4)$.
3. Effect on Higher-Level Hashes: Since $H'_{34}$ changed, any parent nodes hashing $H_{34}$ will also change. For example, the combined hash at the next level involving $H_{34}$ will become a different hash $H'_{1234}$.
4. Changes Reach the Merkle Root: Eventually, the Merkle Root hash changes to a new value $MerkleRoot'$.

**Data Integrity**

- **Tampering Detection:** Because the Merkle Root uniquely summarizes all transactions in the block, any change—even in a single transaction like T3—changes the root hash. This difference signals tampering.
- **Efficient Verification:** Verifiers don't need to check all transactions again but can use a **Merkle Proof** (a path of hashes from the changed leaf up to the root) to confirm integrity quickly.
- **Immutable Chain:** Blocks reference previous blocks' Merkle Roots in their headers, so any tampering in prior blocks breaks the chain's validity, ensuring strong data integrity across the blockchain.

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4. A wallet has 2 ETH initially.Find the final balance of the wallet after the following transactions : Sent 0.4 ETH; Received 1.2 ETH; Paid gas fee of 0.002 ETH.

**Solution:** Balance = 2 − 0.4 + 1.2 − 0.002 = **2.798 ETH**

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5. Given the Initial wallet = 5 ETH; Tx1: Send 1.5 ETH (gas 0.003); Tx2: Receive 0.8 ETH ; Tx3: Send 0.5 ETH (gas 0.002). Compute the Final Balance

**Solution** : Final balance = 5 − 1.5 − 0.003 + 0.8 − 0.5 − 0.002 = **3.795 ETH**

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6. A user with 0.1 BTC sends 0.04 BTC to A and 0.05 BTC to B. The transaction fee is 0.0005 BTC. Find how much change (if any) is returned.

**Solution :** Total sent = 0.04 + 0.05 + 0.0005 = 0.0905 BTC
Change = 0.1 − 0.0905 = **0.0095 BTC**

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7. Alice has 3.2 ETH. She sends 1.1 ETH to Bob and pays gas of 0.002 ETH. Later receives 0.6 ETH from Carol. Find her final balance.

**Solution:** Balance = 3.2 − 1.1 − 0.002 + 0.6 = **2.698 ETH**

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8. A Solidity function execution consumes gas units for Base transaction: 25,000; Data input: (400 bytes × 12 gas each); Storage update: 32,000. If the gas price = 45 Gwei, and 1 Gwei = $10^{-9}$ ETH, find the total gas fee in ETH . *(Assume 1 ETH = $2800.)*

**Solution:**

Total gas = 25,000 + (400×12) + 32,000
= 25,000 + 4,800 + 32,000
= 61,800

Convert gas price to ETH: 45 Gwei = $45 \times 10^{-9}$ ETH per gas

Total gas fee in ETH = $61{,}800 \times 45 \times 10^{-9}$
= $2.781 \times 10^{-3}$ ETH

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9. A simplified hash function outputs values from 0 to 9999. The mining rule: **hash < 500**. If the hash function is **hash = (txid + nonce) mod 10000**, and txid = 7654, find the smallest nonce that satisfies the condition.

**Solution:**

Its a trial & Error Method

- For nonce = 346 :
  - hash = (7654+346) % 10000
  - hash **!<** 500 ⇒ **nonce = 346 is not a golden nonce**
- For nonce = 2346
  - hash = (7654+2346) % 10000 = 0
  - hash = 0 ⇒ **nonce = 2346 is a golden nonce**

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10. A simplified mining algorithm uses **hash=(txid+nonce) mod 10000.** A block is valid iff: **hash < 350.** If **txid = 9675**, determine
(a) **minimum nonce** satisfying the condition;
(b) **resulting hash value**.

**Happy Learning!!!**